A Model Value Transfer Incentive Mechanism for Federated Learning With Smart Contracts in AIoT

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***Abstract*—Introduced by Google in 2016, federated learning (FL) is a distributed machine learning framework to ensure data privacy amid the surge in big data. FL enables secure data sharing without accessing local data. Despite its advantages, it faces challenges due to the limited participation of the data owner. To address this, this article proposes the model value transfer incentive (MVTI) to enhance FL incentives for Artificial Intelligence of Things (AIoT). MVTI allows active participation of data requesters in FL training, addressing limited data owner engagement, and facilitating personalized model construction. The integrated model bail and contribution assessment mechanism ensures fair benefit redistribution. Using smart contracts (SCs) and interplanetary file system (IPFS) enhances security and reliability, ensuring transparent and tamper-resistant execution for secure transactions and data integrity. Our experiments highlight MVTI's superiority in addressing FL incentive challenges for AIoT compared to state-of-the-art baselines on real-world datasets. We also demonstrate the compatibility of multiple gradient protections with incentive mechanisms, especially with gradient compression. The proposed SC-MVTI scheme is resilient and demonstrates the potential to significantly improve the overall efficacy of the FL system within incentive frameworks.**

***Index Terms*—Artificial Intelligence of Things (AIoT), federated learning (FL), incentive mechanism, model value transfer, smart contract (SC).**

Received 4 July 2024; revised 4 September 2024 and 13 September 2024; accepted 20 September 2024. Date of publication 26 September 2024; date of current version 24 January 2025. This work was supported in part by the Yunnan Key Laboratory of Blockchain Application Technology under Grant 202105AG070005, Grant 202305AG340008, and Grant YNB202301; in part by NSFC under Grant 72293583, Grant 72293580, Grant 62476007, and Grant 62271234; in part by the Open Foundation of State Key Laboratory of Networking and Switching Technology (Beijing University of Posts and Telecommunications) under Grant SKLNST-2024-1-06; in part by the Project of Science and Technology Major Project of Yunnan Province under Grant 202302AF080006; in part by the Open Foundation of State Key Laboratory of Public Big Data (Guizhou University) under Grant PBD2022-16; and in part by the Double First-Class Project for Collaborative Innovation Achievements in Disciplines Construction in Heilongjiang Province under Grant LJGXCG2022-054. *(Corresponding author: Kejia Zhang.)*

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Digital Object Identifier 10.1109/JIOT.2024.3468443

I. Introduction

As ARTIFICIAL Intelligence of Things continues to advance rapidly, and big data has emerged as a crucial catalyst for various advancements and innovations [1]. It involves numerous aspects, data collection, storage, processing, and applications. Among them, data sharing is a significant aspect of big data applications, which can promote the sharing and reuse of data resources and improve the utilization and value of data [2]. However, traditional centralized data-sharing methods face challenges (for example, data security, data privacy, and data legitimacy) [3]. To address these issues, federated learning (FL) has emerged as a distributed data-sharing method that can conduct model training and data sharing while ensuring data privacy [4], [5]. Despite practical limitations and challenges, FL has the potential to encourage cross-department and cross-industry data collaboration [6], improving efficiency and security [7] by incorporating technologies, such as blockchain [8], [9], differential privacy (DP) [10], and compression. It has also been widely utilized in sensitive data fields (e.g., healthcare [11] and intelligent transportation [12], [13]), demonstrating its potential to revolutionize various industries.

FL has emerged as a transformative paradigm in machine learning, enabling collaborative model training across decentralized devices while preserving data privacy. Despite its potential, the widespread adoption of FL faces a significant challenge, i.e., Participants' reluctance to share sensitive data [14], [15]. This collaborative approach comes with inherent costs, including the consumption of fundamental resources, such as computational power, network bandwidth, and battery life. Besides, FL is susceptible to malicious attacks by adversaries [16], [17], [18]. Parties may hesitate to participate in FL training without adequate incentives. In this way, providing robust incentives can serve as a solution to this issue by encouraging honest participation and deterring potential

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attackers. By offering rewards for good behavior and implementing penalties for malicious actions, we can create a more secure and trustworthy environment for FL, thereby reducing the likelihood of adversarial attacks. Therefore, incorporating effective incentive mechanisms becomes a cornerstone to encourage participation and foster enthusiasm among data owners [19].

Unfortunately, FL can still be susceptible to malicious attacks by adversaries. These attacks can compromise the privacy of the local models and the global model or affect the accuracy and performance of the model. Participants may also have concerns about potential misuse of their data or models by other parties involved in the FL process. Moreover, participating in FL requires resources, such as computation power and communication bandwidth, for training and sharing the local models. Some parties may not be willing to contribute these resources without proper incentives, as they may not see immediate benefits from participating in the FL process. Therefore, even though FL offers data privacy, incorporating effective incentive mechanisms is crucial to encourage participation and foster enthusiasm among data owners. This is because potential risks and resource requirements might make some parties hesitant to participate without adequate incentives.

A. Our Motivations

To craft compelling incentives for FL participants, a variety of incentive mechanisms have been developed by researchers [19], [20], [21], [22], [23]. These mechanisms aim to effectively motivate FL participants by offering rewards or compensation, thereby ensuring their active engagement in the collaborative training process. Beyond the surface, it is imperative to explore the nuanced interplay among model performance, incentive mechanisms, and the delicate equilibrium needed to navigate the triangle of model performance, incentive design, and data privacy.

1) *(G1.) Model Performance Optimization:* Achieving optimal performance in FL is not only dependent on active participation of participants but also on continuous improvement of the global model. Incentive structures should be meticulously designed to encourage participants not only to contribute data, but also to strive for improved model accuracy and efficiency. Recognizing and rewarding those whose contributions significantly elevate the overall performance of the federated model can be a key driver.
2) *(G2.) Diverse Incentive Structures for Diverse Participants:* FL environments encompass a diverse array of participants, each with unique data characteristics and computing capabilities. Incentive mechanisms must be adaptable and inclusive, recognizing the varied contributions of participants with different types of data and resource capacities. This inclusivity fosters a collaborative ecosystem where all participants feel valued and motivated to contribute, irrespective of their individual circumstances.
3) *(G3.) Incentive Mechanisms and Data Privacy Dilemma:* While incentives drive participation, they must not compromise the privacy of individual data. It is crucial to strike a delicate balance, fostering collaboration without infringing on the participants' privacy. Novel incentive mechanisms must prioritize privacy preservation, employing techniques like DP and secure aggregation to ensure participants feel secure contributing their valuable data to the federated model. However, robust privacy measures can limit information sharing among participants, potentially impeding model convergence. Effective incentive mechanisms should acknowledge and address this tradeoff, providing flexible rewards that consider both the quality of contributions and the level of privacy preservation maintained by participants.

B. Challenges To Be Addressed

FL incentive mechanisms can be broadly categorized into token-based rewards [24], reputation-based trust [22], [25], and game-theoretic incentives [13], [19], [21], [26]. Token-based rewards focus on motivating participants through tokens or cryptocurrencies, particularly suitable for environments that emphasize value exchange. Reputation-based trust aims to establish and maintain trust relationships among participants, which is well suited for scenarios emphasizing cooperation and trust building. Game-theoretic incentives leverage game theory to design rewards and penalties, influencing participant behavior, and addressing dynamics of competition and collaboration. In practical applications, a combination of these mechanisms may be employed to construct a comprehensive and flexible incentive system. Although these methods effectively achieve the first goal, limited attention has been paid to the latter two goals. To address this gap, we highlight the challenges that still need attention in the following way.

1) *(C1.) Personalized Needs:* Current FL incentive mechanisms prioritize global model generalization but often overlook individual client needs, especially when data and tasks differ, leading to poor local performance. Moreover, the 'pay-before-verification' model limits data requester involvement, reducing practical effectiveness. Thus, solving key challenges in FL to address personalized needs is crucial.
2) *(C2.) Training Costs:* Furthermore, data-sharing rewards typically come only from the task initiator (i.e., data requester) and do not consider the model benefits obtained by low-quality data providers from each round of distributed models. Previous approaches have focused mainly on excluding low-quality data providers, missing the opportunity to share training costs with them.
3) *(C3.) Data Privacy:* Since contribution assessment requires verification of model accuracy, previous work is mainly focused on the accuracy of contribution evaluation while overlooking data privacy and security concerns. These problems motivated the following research.

C. Our Contributions

In this article, we resolve the above problems affirmatively and summarize the main contribution below.

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TABLE I
GLOSSARY OF SYMBOLS

N	The number of data providers
R_0	The data requester
V_0	The validation dataset
T_0	The test dataset
D_i	The local dataset
P_i	The training quality of data provider
E	The round number of SGD
W_t	The model parameters
ΔW_t	The updated gradients
$\hat{c}_t$	The model sharing price
b_t^k	The model bids from data providers for $k = \{0, 1, \cdots, n-1\}$
w_t	The model after aggregating all gradients
w_t^{-j}	The model after aggregating gradients except for those from the data provider j
$L(*, V_0)$	The loss function evaluated in the validation data set V_0 of the data requester.
I_j	The importance of the gradient of the data provider j
π_t^j	The contribution of the data provider j in the t-th round

1) We introduce model value transfer incentive (MVTI), a novel FL incentive mechanism for Artificial Intelligence of Things (AIoT). MVTI enables active engagement of data requesters in FL training, facilitating personalized model construction to meet specific task requirements. We also integrate a model bail and contribution assessment mechanism that ensures fair redistribution of training benefits, encouraging collaboration among participants, even those with lower quality data contributions.
2) We leverage smart contracts (SCs) and the interplanetary file system (IPFS) technique to enhance the security and reliability of MVTI in AIoT. This combination ensures transparent and tamper-resistant execution of the incentive mechanism, providing a robust foundation for secure transactions and data integrity in the FL ecosystem.
3) We conducted extensive experiments to compare the convergence speed and evaluate the effectiveness of the proposed MVTI compared to state-of-the-art baselines in real-world AIoT datasets. Furthermore, gradient leakage attack experiments were conducted to test the compatibility of mainstream privacy protection strategies, and gradient compression (GC) demonstrated good compatibility.

These findings show the effectiveness and superiority of our SC-MVTI scheme in addressing the challenges associated with FL incentive mechanisms. Our approach demonstrates a compelling capability to navigate and overcome hurdles, showcasing its potential as a robust solution for enhancing the efficacy of FL systems in the context of incentive frameworks in AIoT.

D. Outline of This Article

The organization of this article is as follows for reviewers and readers. We review the literature in Section II. Section III covers the preliminary knowledge. After that, in Section IV, we present our proposed incentive mechanism. Our comprehensive performance evaluation is detailed in Section V. In summary, we conclude this article in Section VI. We summarize the terms used in the context and their description in Table I.

II. RELATED WORKS

To enhance the participation and motivation of data owners in FL data sharing, researchers have developed various incentive mechanisms. These mechanisms include reputation-based systems, payment transfer methods, auction theory, contract theory, and Shapley value (SV). As introduced above, the approaches can generally be categorized into three main types.

A. Token-Based Incentive Mechanisms

These mechanisms are a federated payment incentive method that uses digital cryptocurrencies or tokens to incentivize participants to share data and models, compensate them for sharing resource consumption, and reward them for their contribution as participants in the global model [27]. Assessing the contribution of participating parties has always been a research focus for incentive algorithms in FL. Traditional FL algorithms evaluate the contribution based on the number of participant samples, which may lead to false reporting and arbitrage opportunities. Therefore, the researchers have introduced the SV to evaluate the quality of the data from the participating parties [28], [29]. Due to the high-computational cost of computing SV, Ghorbani and Zou [30] proposed an approximate SV approach called TMC-Shapley, which uses Monte Carlo sampling techniques to approximate the true SV. Shi et al. [31] proposed the COS-SIM method, which approximates SV using cosine similarity.

B. Reputation-Based Trust Mechanisms

These mechanisms are a type of federated incentive method that evaluates the reputation and contribution of participants to determine their trust and rewards in FL based on their past performance [32]. Zhao et al. [33] proposed a reputation-based FL incentive system that uses blockchain technology. Initially, all participants are assigned equal reputation values on the blockchain. However, the importance of each participant is dynamically adjusted based on the value of the model

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updates they provide. Participants with higher reputations have a higher likelihood of being selected, while those with lower reputations may be excluded. This approach aims to encourage honest participation and encourage the contribution of high-quality model updates. Another solution proposed by ur Rehman et al. [34] incorporates both participant importance records and node-to-node evaluation records. The model aggregation and reputation update process is automated through smart contracts (SCs), ensuring the reliability of model aggregation and real-time updates of reputation values. This automation promotes honest participation in FL training and improves the overall efficiency and effectiveness of the incentive mechanism. However, it should be noted that both of these works lack a comprehensive quality evaluation mechanism. Additionally, the participant rating mechanism used in these systems is unique and vulnerable to malicious attacks.

C. Game-Based Incentive Mechanisms

These mechanisms are methods to motivate participants to participate in FL and improve their willingness to cooperate through strategies and benefits in game theory [35]. This approach typically models the behavior of participants as a game process and guides them in making reasonable decisions by developing appropriate incentive strategies. Lim et al. [36] used contract theory to establish data-sharing cooperation between clients and users, making the data-sharing process more fair and transparent. Furthermore, considering the differences among different clients, the authors proposed a hierarchical incentive model for reward allocation. Zhan and Zhang [20] proposed a game-based incentive mechanism based on game theory and deep reinforcement learning. The server distributes tasks to multiple client devices for model training and rewards them based on their training data volume and model contribution.

D. Our Proposed Scheme for AIoT

The SC-MVTI scheme is classified as a token-based incentive mechanism. We emphasize the value of each round of task models and introduce a model bail and contribution evaluation mechanism to reallocate the model revenue of data providers. After that, we use blockchain to establish a trusted and decentralized execution environment for incentive mechanisms. SCs, acting as automated contracts with predefined logic, can define rules and conditions for incentive mechanisms. The transparency of the blockchain enables transactions and SC execution records to be visible, facilitating verification of their legitimacy. This transparency also improves the auditability and supervision of the incentive execution process, mitigating the risks of fraud and misconduct. In addition, the compatibility of the incentive mechanism with data leakage issues is also taken into account. GC is used to protect data privacy while reducing communication and model storage costs.

III. Problem Formulation

A. Problem Definition

Horizontal FL (HFL) is a machine learning technique where multiple devices collaborate to train a model without sharing their data with each other [37]. It involves a data requester and N data providers, whose roles remain consistent within the same task, but may vary between different tasks. The data requester, denoted R_0, has a validation dataset V_0 to evaluate the quality of the training of the data provider P_i, as well as a test dataset T_0 to verify the performance of the trained model. Data providers assist the data requester in FL model training by utilizing their local dataset D_i. This process involves multiple rounds of iterative optimizations. In each training round, the random model distribution is modified to a predetermined model distribution to prevent free-riding. Upon obtaining the model parameters W_t, the data providers optimize them using their local data, and after E rounds of stochastic gradient descent (SGD), they send the updated gradients ΔW_t to the data requester. The data requester calculates the contributions of data providers based on the quality of their gradients and subsequently distributes the reward and model bail accordingly.

B. System Architecture

As shown in Fig. 1, the SC-MVTI framework consists of data providers, a data requester, IPFS [38] model storage, and a MVTI-based SC. The data requester plays a crucial role in the framework, as it not only initiates the FL task, but also performs gradient quality detection, model aggregation, and contribution evaluation. Unlike previous schemes, which treated data requesters as passive participants, our SC-MVTI framework empowers them to actively participate in model training and provide feedback for customized demands. To prevent the latest model from being lost and to incentivize excellent data providers, we introduce model collateral settings, ensuring that the efforts put into training the model are not in vain. It achieves a tradeoff for training benefits among data providers. To protect data privacy, we use Top$k$ sparsification to compress gradients, thus preventing data leakage and reducing communication and storage costs. Eventually, our SC-MVTI mechanism incorporates a SC based on MVTI, which is deployed on the blockchain. It enables transparent and automatic management of data-sharing tasks, reducing disputes and default risks between data providers and requesters, and improving the overall reliability.

The SC-MVTI workflow can be described as follows.

1) To initiate an FL assistance task, a data requester creates a task ID and specifies the model, dataset format, contract address, and other relevant details. Subsequently, the data requester publishes the FL assistance training message to the data sharing community. Upon receiving a message, the data provider evaluates its own situation and decides whether to participate in a task.
2) After a data requester has trained the model locally, the current model parameters are uploaded to IPFS for storage. The IPFS address and model bail are then recorded and managed in the MVTI SC. To incentivize data providers, a data requester pays the current round of rewards to the SC. It ensures that data providers are compensated for their contributions to the FL assistance task.

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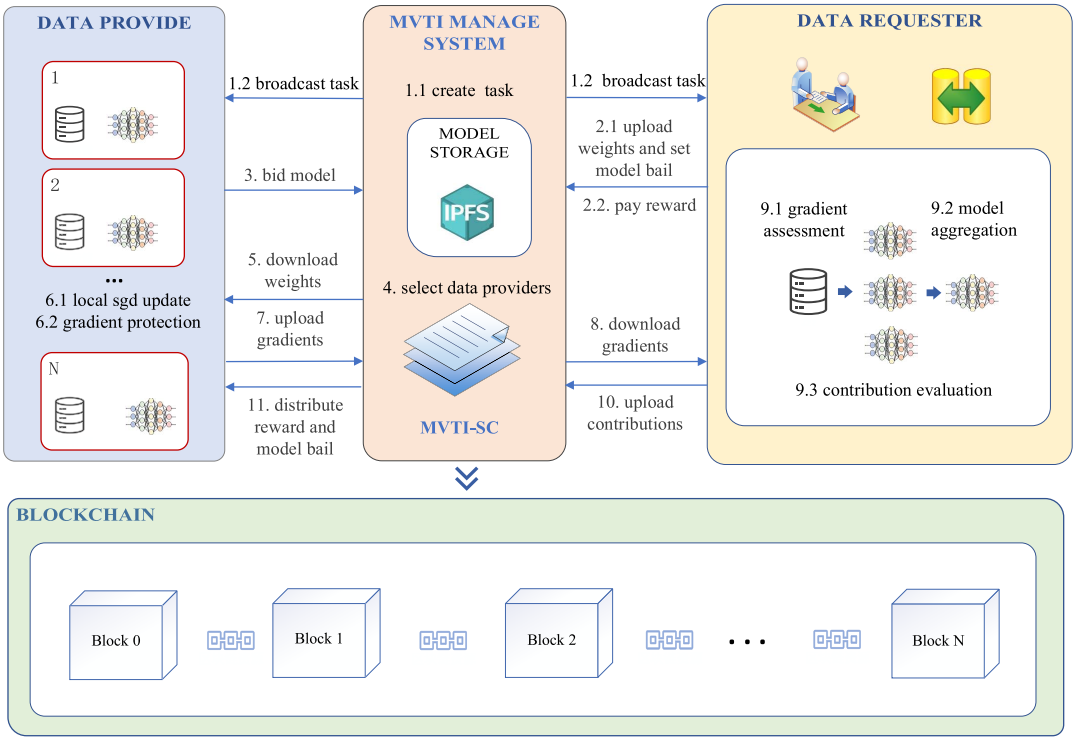


Fig. 1. Our Proposed SC-MVTI Framework for AIoT.

3) The data provider then pays the bidding price for the current round of model to the MVTI SC.
4) After receiving bids from data providers, the MVTI SC sorts them from lowest to highest and selects the bid that satisfies the minimum splitting model bail after deduction. The selected data provider is then authorized to access the current round model parameters stored in IPFS. The remaining bidding price is refunded to the other data providers. This ensures that data providers are fairly compensated for their contributions to the FL assistance task, while also preventing unnecessary waste of resources.
5) After that, the data provider downloads the latest model parameters based on the IPFS address.
6) The data provider leverages local data to optimize and update the model parameters. Once local training is completed, the gradient is calculated and compressed for efficient transmission.
7) The data provider uploads above gradient to IPFS and submits IPFS address to the MVTI SC.
8) The data requester obtains current round gradients from the SC until all data providers have submitted or the task has timed out.
9) The data requester assesses the quality of the gradients by analyzing the local validation dataset and then combines the next round of model parameters, taking into account the weights assigned to each gradient. Finally, the contribution of each data provider is determined on the basis of the impact of their respective gradients.
10) The data requester uploads the data provider's contributions to the MVTI SC.
11) After the data providers submit their gradients, the MVTI SC divides reward equally among them. The model bail is then distributed to data providers based on their contributions, achieving a secondary distribution of benefits derived from the model.
12) Repeat the above steps until the model converges or the predefined round is reached.

IV. Our Proposed Incentive Mechanism—SC-MVTI

In this section, we describe our proposed SC-MVTI mechanism in detail, which consists of five procedures.

A. Data Provider Selection

In vanilla FL, data providers are required to participate in model training without any conditions. However, there is a significant disparity in the benefits received by different data providers, which can be attributed to two primary factors.

1) The first factor that contributes to the disparity in benefits among data providers is the varying quality and quantity of data they contribute. Some providers may possess a large volume of high-quality data, while others may have limited or lower quality data. This difference in data quality can result in some providers making a more significant contribution to model improvements during the training process, while others may make a relatively smaller contribution.

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2) The second factor is the variation in the resources and efforts invested during the training process. Some contributors may allocate more computational resources and time to participate in the training, while others may contribute fewer resources. It leads to an imbalance between resource input and training benefits, with some providers receiving a more significant benefit from the training process than others.

To overcome these challenges, researchers have proposed various FL incentive mechanisms. Although these mechanisms offer additional incentives for high-quality data providers, they often overlook the value of the task model itself, which can lead to potential model arbitrage. To overcome this limitation, we present a novel model bail mechanism, in which data providers can only access the latest model after paying a specific model bail to prevent loss of training results. Subsequently, the model bail is redistributed based on the contributions of the data providers. On the basis of the aforementioned description, our SC-MVTI model can be defined as follows.

1) Assume that the data requester sends a message to the data sharing community, seeking assistance for the current round of the FL task, with a reward of B_t and a model bail of C_t. Upon receiving a message, data providers will evaluate whether to participate in the current training task, considering several (e.g., data factor, computing resources, model value, and account balance). If they choose to participate, they will provide a model bid of b_j^t for the latest model; Otherwise, they will ignore this message.
2) Once all model bids are collected from the data providers, the data providers for the current round are selected, and the model share price $\hat{c}_t$ is calculated. After deducting the model sharing price from the total reward, the remaining amount is returned to the data providers. As the model bail for each round is limited, the following equation should be satisfied:

$$\sum_j \hat{c}_t x_j = C_t, b_j^t - c_t \geq 0 \tag{1}$$

where $x_j \in \{1, 0\}$ represents whether a data provider is selected, with 1 indicating selection and 0 indicating non-selection. Given that the data requester has set the model benchmark, the objective is to encourage as many data providers as possible to participate in FL. It can enhance the generalization of the model by increasing the diversity of samples used for training.

First, sort the model bids from the data providers in Descending order: $b_0^t \geq b_1^t \geq \cdots \geq b_{n-1}^t$, where n represents the total number of data providers that can provide FL assistance services. At the beginning of the FL task, the data providers are invited by the data requester and then recommended by the system. Second, the system selects data providers who satisfy the minimum splitting model bail according to (2) and uses (3) to calculate a model sharing price in the tth round

$$b_k^t \geq \frac{c_t}{n-k} \tag{2}$$

Algorithm 1: MVTI Data Provider Selection

Input: C_t, $\{b_0^t, b_1^t, \ldots, b_{n-1}^t\}$
Output: $\hat{c}_t$, $\{P_0^t, P_1^t, \ldots, P_k^t\}$

Sort the bids $b_0^t, b_1^t, \ldots, b_{n-1}^t$ in descending order
Initialize $k \leftarrow 0$, $\hat{c}_t \leftarrow C_t$
while $k < n$ **do**
  if $b_k^t \geq \frac{C_t}{n-k}$ **then**
    $\hat{c}_t \leftarrow \frac{C_t}{n-k}$ **break**
  end
  $k \leftarrow k + 1$
end
if $k = n$ **then**
  Return $\hat{c}_t$, $\{\varnothing\}$
end
Return $\hat{c}_t$, $\{P_0^t, P_1^t, \ldots, P_k^t\}$

$$\hat{c}_t = \frac{C_t}{n-k}. \tag{3}$$

The detailed process for selecting data providers is shown in Algorithm 1.

This algorithm selects data providers greedily, prioritizing those with lower model bids to encourage participation and enhance training data diversity. To address the risk of malicious data providers potentially stealing training results—since the model can be copied infinitely once released—the data requester sets a model bail for each training round, which is redistributed to the participating providers. This approach helps rebalance the benefits among data providers. The time complexity is $O(n \log n)$, ensuring efficient selection even with a large number of bids.

B. Gradient Quality Assessment

In situations with asymmetric information in AIoT, it is challenging for the data requester to assess the quality of the data provided by data providers. Vanilla FL often relies on self-reported sample volumes, which may be unreliable due to differences in data distribution and sample noise among providers. Solely depending on self-disclosure is inadequate. A more intuitive method involves evaluating the quality of data providers based on the loss of their gradients on the data requester's validation dataset. However, a single gradient might not significantly improve the performance of the global model, and it is the aggregation of multiple gradients that leads to better outcomes. Another approach is to evaluate gradient quality based on distance or angular differences between gradients. Nonetheless, this can lead to overly convergent gradients, potentially compressing the parameter optimization space of the global model. Xu et al. [39] noted that in non-IID scenarios, local models frequently get trapped in saddle points, resulting in degraded performance when gradients are passed to the global model.

Inspired by the work of Zhang et al. [25], the marginal utility of gradient aggregation is used to evaluate the quality of gradients. This method yields relatively stable results and

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can be expressed as follows:

$$w_t = w_{t-1} + \frac{1}{K}\sum_{i=0}^{K-1} \Delta w_i^t \tag{4}$$

$$w_t^{-j} = w_{t-1} + \frac{1}{K-1}\sum_{i\neq j} \Delta w_i^t \tag{5}$$

where w_t denotes the model after aggregating all gradients and w_t^{-j} denotes the model after aggregating all gradients except those from data provider j. The model has p parameters. The initial algorithm has a time complexity of $O(K^2p)$ due to the need to recompute the aggregated model for each data provider. To improve efficiency, especially for scenarios involving a large number of devices in AIoT, we optimize the algorithm by precomputing the sum of all gradients. This allows us to adjust the model excluding each data provider without recomputing the entire sum from scratch.

1) *Precompute Sum of All Gradients:* Compute the total sum S of gradients once

$$S = \sum_{i=0}^{K-1} \Delta w_i^t. \tag{6}$$

2) *Efficient Model Updates:* Use S to compute w_t

$$w_t = w_{t-1} + \frac{S}{K}. \tag{7}$$

3) *Adjust for Exclusions:* For each data provider j, compute w_t^{-j} using

$$w_t^{-j} = w_{t-1} + \frac{S - \Delta w_j^t}{K-1}. \tag{8}$$

By employing this technique, the time complexity is reduced from $O(K^2p)$ to $O(Kp)$, making the algorithm more scalable and efficient for a large number of data providers or devices.

The marginal utility of a gradient from a data provider can be expressed as

$$D_t^j = L\left(w_t^{-j}, V_0\right) - L(w_t, V_0) \tag{9}$$

where $L(*, V_0)$ represents the loss function evaluated on the validation dataset V_0 of the data requester.

The importance of the gradient from provider j is quantified by the score I_j^t, calculated using the Gompertz function

$$I_j^t = ae^{be^{cD_t^j}} \tag{10}$$

where a, b, and c are hyperparameters that control the shape of the Gompertz function, and D_t^j is the change in loss due to the gradient from provider j. In this article, a Gompertz function with hyperparameters $a = 1$, $b = -4$, and $c = 6$ is used. The distribution of this function is shown in Fig. 2.

The importance score I_j^t adjusts each gradient's weight in the model aggregation process, ensuring that more relevant gradients have a greater impact on model updates. This leads to more effective FL training in AIoT.

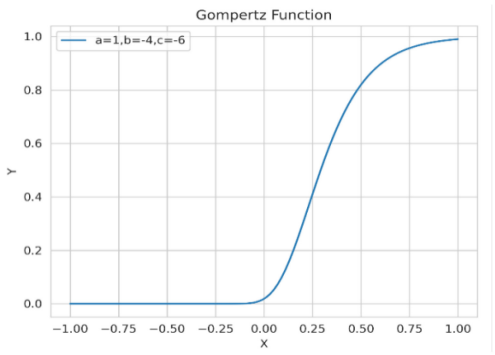


Fig. 2. Gompertz function.

C. Model Aggregation With Historical Gradient Projection

The previous section presented a method for evaluating the gradient of a data provider. In this section, we use the gradient's importance score as a weight, replacing the sample volume used in traditional FL. The model aggregation update equations are

$$\alpha_t^j = \frac{I_t^j}{\sum_{i=0}^{K-1} I_t^i} \tag{11}$$

$$w_t = w_{t-1} + \sum_{j=0}^{K-1} \alpha_t^j \Delta w_t^j. \tag{12}$$

The basic aggregation method has been outlined. Let q_0 be the size of the validation dataset and p be the number of model parameters. Evaluating the loss function for each data provider has a complexity of $O(q_0p)$. As the evaluation is performed for each provider, the total complexity is $O(Kq_0p)$. Computing the importance score I_j^t using the Gompertz function has a complexity of $O(1)$ per provider, leading to an overall complexity of $O(K)$. Normalizing these importance values to compute the weight α_t^j involves both summation and division, each requiring $O(K)$ time. Thus, the complexity for weight calculation is also $O(K)$. The model update equation involves a weighted sum of the gradients from all providers, which requires $O(Kp)$ operations.

Many studies have demonstrated that leveraging historical gradients can accelerate model convergence and broaden the optimization space for model parameters in FL. This section proposes a momentum optimization strategy for aggregated gradients. Unlike previous approaches [40], [41], where historical gradients are managed by individual data providers, our method centralizes the management and maintenance of historical gradients with the data requester. Furthermore, instead of aggregating the gradients from various data providers, our approach modifies and enhances each provider's own gradients individually. To address potential absences of data providers during FL training in AIoT environments, historical gradients are integrated with the most recent gradients rather than being reset. This approach encourages data providers to maintain a consistent optimization direction. A schematic of this process is shown in Fig. 3.

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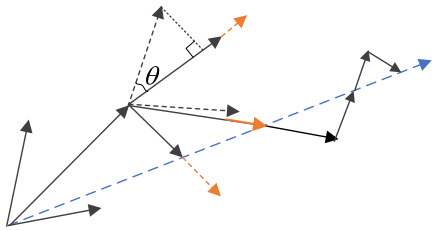


Fig. 3. Projection of historical gradients.

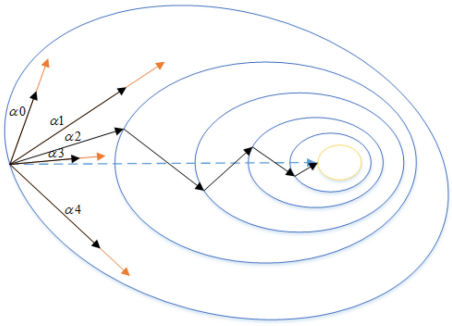


Fig. 4. Contribution of data provider gradients to global model updates.

The gradient update equation for data providers can be expressed as follows:

$$\Delta w_t^j = \Delta w_t^j + \beta \text{proj}\left(m_{t-1}^j, \Delta w_t^j\right) \tag{13}$$

$$m_t^j = \Delta w_t^j \tag{14}$$

where the historical gradients are projected onto the latest gradient and accumulated with the current gradient before aggregation. Note that gradient quality assessment is performed using the original gradients from the data providers, instead of the enhanced ones. The process of accumulating the projected historical gradients with the current gradients has a complexity of $O(p)$ per provider, leading to a total complexity of $O(Kp)$ across all providers.

D. Contribution Evaluation and Reward Allocation

Contribution evaluation is the most critical step in incentive mechanism design. Traditional SV methods [30], [42], [43] are computationally expensive, while SV approximation methods based on sampling or cosine similarity are sensitive to data and lack stability. Therefore, this article adopts a contribution evaluation method similar to that in [31], which uses the projection of the data provider's gradient on the global update as a basis, combined with the relative importance of the data provider's gradient. The expression is

$$\pi_t^j = \alpha_t^j \cdot \text{proj}\left(\Delta w_t^j, \Delta w_t\right) \tag{15}$$

where π_t^j represents the contribution of the data provider j in the tth round. It is worth noting that the gradient mentioned here refers to the gradient after undergoing historical gradient correction, and data providers who have consistent and positive optimization directions will have an advantage in contribution evaluation. The illustration of this process is shown in Fig. 4.

In terms of reward distribution, the data requester will upload the data providers' contribution to the MVTI SC, which will allocate the current round model bail based on the relative contribution ratio of the data providers. The value of the previous round model will be transferred to high-quality data providers, while indirectly collecting usage fees from low-quality data providers. The current round reward will be evenly distributed among data providers who upload gradients, thereby fully motivating them to upload gradients instead of only purchasing models. Finally, the actual income of the data provider j in the tth round can be expressed as

$$p_j^t = \frac{C_t \cdot \max\{\pi_j^t, 0\}}{\sum_{j=0}^{K-1} \max\{\pi_j^t, 0\}} + \frac{B_t}{K-1}. \tag{16}$$

Finally, we analyze the complexity of contribution evaluation and reward distribution. The computation of the contribution π_t^j involves projecting the gradient Δw_t^j onto the global update Δw_t. This projection operation has a complexity of $O(p)$ per data provider. Consequently, when applied to all K providers, the total complexity for contribution evaluation is $O(Kp)$. In addition, the reward distribution process involves summing the contributions and calculating the relative shares, both of which require $O(K)$ operations. Therefore, the overall complexity is dominated by the contribution evaluation step, resulting in $O(Kp)$.

In a nutshell, the dominant term in the complexity is $O(Kq_0p)$, which arises from evaluating the loss function across all providers. In comparison to the traditional FedAvg method, which has a complexity of $O(Kp)$, this approach introduces additional computational overhead, particularly when q_0 is small. Nevertheless, this added complexity improves the model's reliability, fairness, and robustness in more complex FL environments.

E. MVTI Smart Contract

In any data collaboration, the rights and interests of the data providers must be protected. In addition to a fair and reasonable incentive mechanism, a secure and trusted execution environment is also necessary. In the MVTI model, the data requester leads the FL training, and to prevent the data requester from withdrawing or underpaying the rewards, the SC manages the task rewards and model bails. The data requester only affects the allocation of model bails through its contribution, rather than directly participating in payment, which allows the data requester to focus on improving the model. To achieve this, we implemented the MVTI model based on SCs and deployed it on the blockchain to decentralize the maintenance of the data-sharing cooperation at the same time, the entire transaction record is recorded on the blockchain, effectively preventing third-party organizations from maliciously tampering with data transactions. The basic functions of SC-MVTI are shown in Fig. 5.

This SC mainly maintains variables, such as data requester, data providers, model parameters, gradients, and model bails and task rewards. In addition, in addition to constructor and destructor, it also includes nine public functions as below.

1) *Register():* The data provider registration interface.
2) *UploadWeight():* The data requester uploads the IPFS address of the current round of model parameters through this function, pays the task reward, and sets the model bail.

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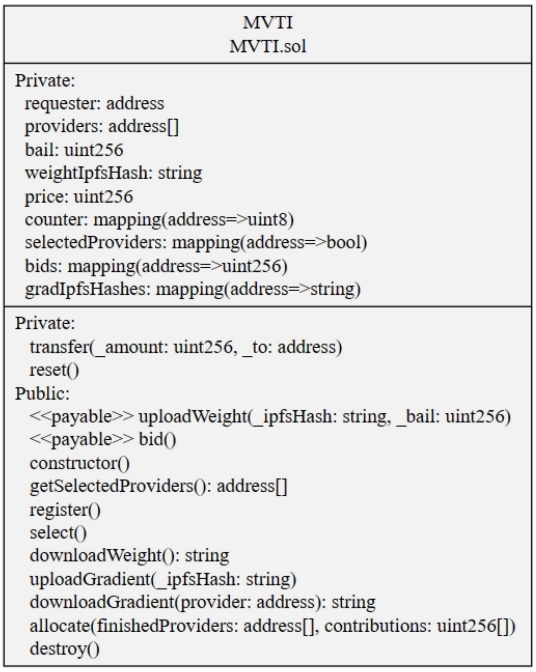


Fig. 5. UML class diagram of SC-MVTI.

3) *Bid():* The data provider pays a certain fee as a model bail for current round of model through this function.
4) *Select():* The data requester or system automatically triggers the selection of the data provider for the current round.
5) *Allocate():* The data requester is to report the contribution of the data provider in the current round.
6) The rest of the four functions are used to upload and download gradients, download model weights, and obtain the selected data providers for the current round.

In FL managing a large volume of model parameters can incur significant gas fees if stored directly in SCs. For example, storing 50MB of model parameters could cost 1 ETH. Additionally, SCs may struggle with accurate floating-point arithmetic, affecting model precision. In AIoT environments with limited computing and storage capabilities, using dedicated storage middleware is essential. To mitigate these issues, we propose using the IPFS to store intermediate model results, while SCs handle the management of IPFS addresses. This approach reduces SC interaction costs and improves storage efficiency, stability, and scalability.

V. Performance Evaluation and Comparison

We evaluate the performance of MVTI scheme in practice to demonstrate its efficiency for AIoT. To begin with, we implement several popular algorithms in Python language using different frameworks, such as PyTorch, Ray, and Brownie. The hardware environment is accomplished by 11th Gen Intel Core i5-11400F CPU with 32-GB RAM and GeForce RTX 3060 under Ubuntu 20.04 operating system. Following the blueprint experiments designed in [31], we adopt four different partitioned MNIST testing datasets. Additionally, we leverage the EMNIST and CIFAR-10 datasets, which are also split similarly, to provide a broader perspective on the performance of each algorithm.

A. Experiment Setup

To assess the effectiveness of the MVTI algorithm in AIoT scenario, we followed the experimental framework proposed by Shi et al. [31]. We utilized three datasets, i.e., MNIST, EMNIST, and CIFAR-10, each of which was partitioned into four different configurations. This approach led to the creation of twelve distinct FL scenarios for comprehensive evaluation.

1) Datasets and Models: This part details the three datasets and models used in AIoT simulations for FL experiments.

1) *MNIST (Handwritten Digit Recognition in Smart Devices):* In this part, MINIST dataset is used to describe the smart devices in AIoT.
   a) *Dataset:* MNIST is a widely used dataset of handwritten digits, containing 60 000 training images and 10 000 testing images of digits ranging from 0 to 9, with each image being 28 × 28 pixels in size.
   b) *AIoT Context:* Imagine this dataset being used in an AIoT environment where smart devices, such as tablets or digital pens, recognize handwritten inputs for digit recognition. This application is relevant for tasks like digitizing handwritten notes or enabling secure numeric input in IoT devices.
2) *EMNIST (Character Recognition in IoT Devices):* As an extension of MNIST, EMNIST dataset is introduced for describing the AIoT scenarios in smart home or office devices.
   a) *Dataset:* EMNIST extends the MNIST dataset, by including more handwritten characters from various writers, particularly using the letters partition to validate multiclass classification scenarios.
   b) *AIoT Context:* This dataset simulates AIoT scenarios in smart home or office devices that require the recognition of a broader range of handwritten characters, such as smart security systems for personalized authentication, or IoT-enabled educational tools that adapt to different handwriting styles.
3) *CIFAR-10 (Object Detection in IoT Cameras):* CIFAR-10 is suited for simulating the AIoT applications in smart surveillance or autonomous systems.
   a) *Dataset:* CIFAR-10 is a dataset consisting of 60 000 32 × 32 color images across 10 different classes, including objects like airplanes, cars, birds, cats, and more.
   b) *AIoT Context:* This dataset is suited for simulating AIoT applications in smart surveillance or autonomous systems where IoT-enabled cameras detect and classify objects in various environments, such as traffic monitoring systems, smart agriculture for identifying different animals or machinery, or home security cameras, that recognize different objects and trigger appropriate actions.

For a thorough evaluation of FL algorithms, we extracted 10 000 samples from each dataset and divided them into four distinct setups to explore different FL scenarios.

1) *Uniform (UNI):* Data was split into 10 equal parts, with each provider receiving one part across the three datasets.

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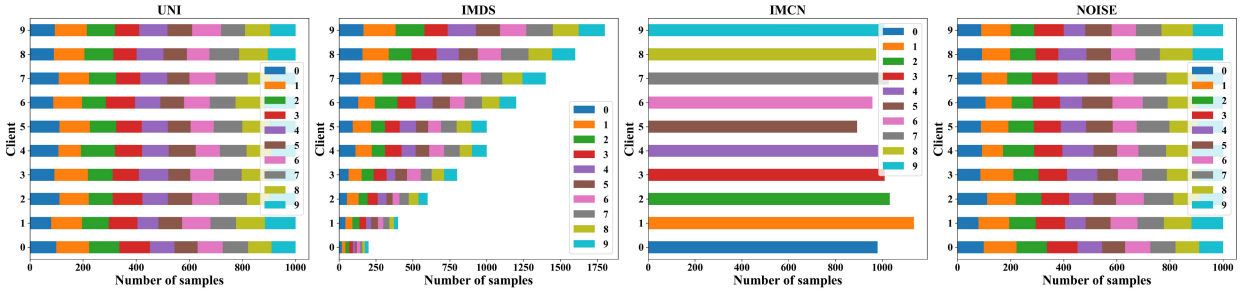


Fig. 6. Schematic of data provider categories distribution on MNIST.

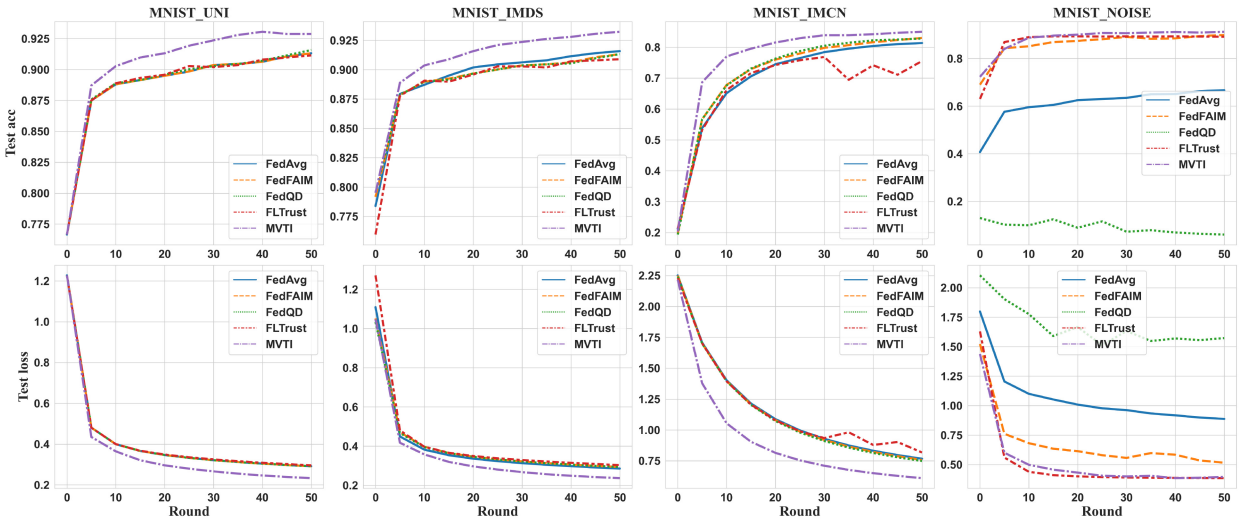


Fig. 7. Test curves of baselines under four scenarios on MNIST.

2) *Imbalanced Dataset Size (IMDS):* Data was partitioned into ten parts with varying sample sizes: 200, 400, 600, 800, 1000 (twice), 1200, 1400, 1600, and 1800. Each provider received a different amount of data across the datasets.
3) *Imbalanced Class Number (IMCN):* Data was split by class. For MNIST, each provider received one class. For EMNIST, providers received overlapping groups of nine classes. For CIFAR-10, providers received overlapping groups of three classes.
4) *Label Noise (NOISE):* Data was divided into 10 equal parts, with labels mixed up by 0% to 90% in steps of 10% for each part. This setup tests the system's robustness against varying levels of label noise across the datasets.

We randomly selected 500 samples from the training dataset for validation and designated 1000 samples as the test dataset for the data requester. The class distribution among data providers is illustrated in Fig. 6. Different neural networks were used for each dataset: 1) an MLP for MNIST; 2) a CNN with two convolutional layers for EMNIST; and 3) a CNN with three convolutional layers for CIFAR-10. These datasets and models enable us to evaluate the MVTI algorithm's robustness and adaptability across various AIoT use cases, offering insights into its performance in real-world FL environments. This part includes four aggregation algorithm for FL experiments.

2) Baselines:
1) *FedAvg:* A weighted model aggregation algorithm based on sample volume [4]. As this algorithm does not consider incentive problems, for the sake of comparison, this article distributes rewards based on the sample volume of each data provider.
2) *FLTrust:* A Byzantine fault-tolerant aggregation algorithm with a self-built trust root [44], which assigns a trust score to each local model to prevent malicious participants from compromising the global model. This algorithm also does not consider incentive problems, so this article distributes rewards based on the trust score of each data provider.
3) *FedQD:* A weighted model aggregation algorithm based on local model loss [25]. Additionally, this article designs a contribution evaluation algorithm based on gradient projection and uses it to distribute rewards to data providers.
4) *FedFAIM:* A weighted model aggregation algorithm based on merge loss [31]. By compared the contribution of each local model to the global model loss, its importance of each participant is evaluated, and rewards are allocated based on participant's importance and gradients.

3) Evaluation Metrics and Hyperparameters: We evaluate FL incentive algorithms based on three main criteria: overall model performance, correctness of contribution, and fairness

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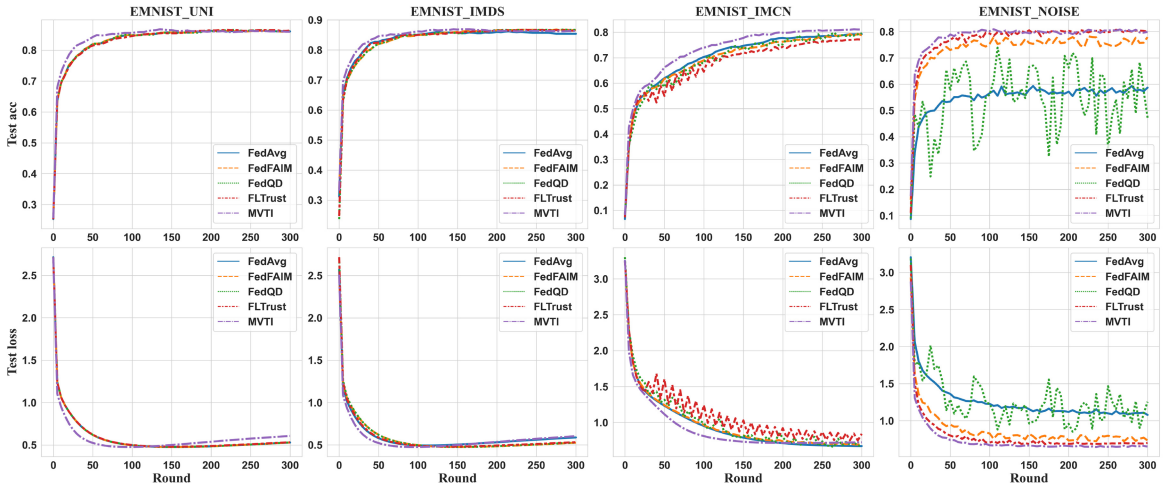


Fig. 8. Test curves of baselines under four scenarios on EMNIST.

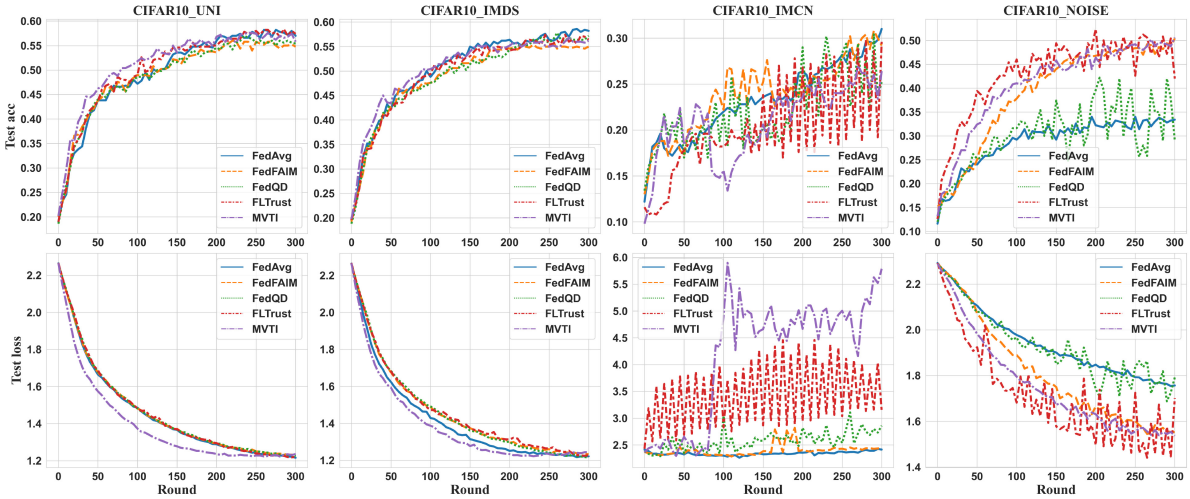


Fig. 9. Test curves of baselines under four scenarios on CIFAR10.

of distribution. Model convergence is primarily evaluated using the top-1 classification accuracy curves in the test dataset provided by the data requester. The correctness is compared by assessing the relative quality of the training samples provided by different data providers in various scenarios. Distribution fairness is evaluated by examining whether the rewards received by the data providers align with their respective contributions. In the global FL setting, with 10 data providers participating, each device performs 5 local SGD iterations per round, using a batch size of 32 and a learning rate of 0.01. The fixed task reward is 2, the model bail is 3, and the default account balance for data providers is 0. The experiments are conducted over 50 training rounds for MNIST and 300 training rounds for EMNIST and CIFAR-10. For the respective methods: FedQD has a base score of 1.0, FedFAIM uses a merge loss threshold of -0.01, FLTrust operates with the gradient regularization option turned off, and MVTI employs a historical gradient projection coefficient of 0.6.

B. Effects of Our Proposed Algorithm

1) Model Convergence Speed: As depicted in Figs. 7–9, MVTI demonstrates excellent performance in most scenarios, primarily due to its use of the historical gradient projection acceleration technique, which leverages the residual heat of previous gradient updates. We provide a detailed analysis.

1) *UNI Scenario:* Across the three datasets (MNIST, EMNIST, and CIFAR-10), the performance of various baselines, such as FedAvg, FedFAIM, FedQD, and FLTrust, is relatively comparable. Notably, MVTI stands out due to its distinct advantages. MVTI demonstrates faster convergence, better stability, and lower loss.
2) *IMDS Scenario:* In this scenario, FedAvg performs relatively well due to its reliance on sample volume for gradient quality. MVTI, however, achieves the same level of performance through a reasonable gradient quality assessment method.
3) *IMCN Scenario:* Here, each data provider possesses only a few classes of samples, which means the gradient from any single data provider can potentially degrade the overall model performance. This is particularly evident in scenarios like CIFAR-10, where interclass differences are significant. However, through aggregation, this situation is reversed. This also explains the decline in FLTrust's performance in later stages, highlighting the need for strengthened cooperation in FL.

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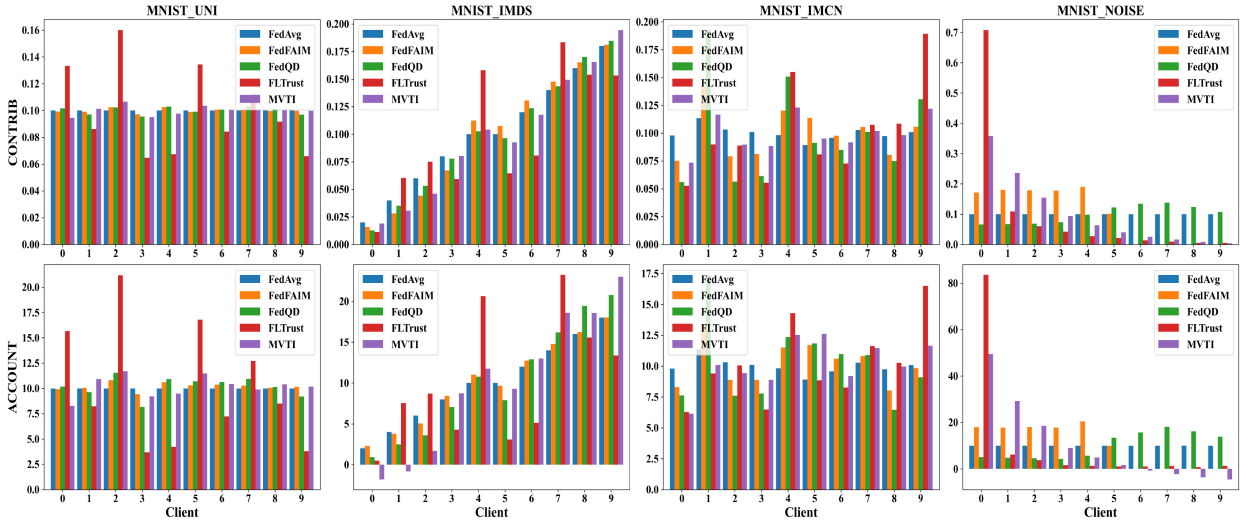


Fig. 10. Contribution and benefits of baselines under four scenarios on MNIST.

4) *NOISE Scenario:* In this scenario, if gradient quality is not checked, gradients from data providers with incorrect labels can significantly disrupt global model training. FLTrust, FedFAIM, MVTI, and FedQD assist the data requester in quickly identifying these problematic gradients by establishing a trusted root with a high-quality dataset. It is important to note that although FedQD also employs gradient detection technology, its stability is insufficient under extreme conditions.

Overall, the superior performance of the MVTI algorithm in FL settings makes it a strong choice for scenarios with varying data distributions and noise levels.

2) Correctness and Fairness: To verify the correctness and fairness of the incentive algorithm, we compared the contribution values and income in four scenarios of the MNIST dataset (see Fig. 10). In the UNI scenario, except for FLTrust, all algorithms correctly evaluated the actual data quality of the data providers. In the IMDS scenario, all algorithms did so as well, but MVTI transferred fees from low-contribution to high-contribution data providers. In the IMCN scenario, data quality was relatively balanced, resulting in no significant difference in contribution and income. In the NOISE scenario, MVTI demonstrated more accurate data quality recognition. The key difference between MVTI and previous incentive schemes is that the economic benefit for data providers can be negative, meaning they might lose money by participating in FL training. Traditionally, data providers are unlikely to participate due to overlooking the performance improvement benefits of FL. MVTI addresses this by transferring model revenue in the form of a model bail, overcoming the issue of "high contribution and low income."

3) Data Privacy: The distributed training has gained widespread adoption in large-scale machine learning. Historically, it was widely accepted that gradient information could not be employed to reverse engineer the original data. However, Zhu et al. [45] introduced an attack called deep gradient leakage (DGL), based on generative adversarial networks, that iteratively recovers user training data from gradients. This challenges the conventional belief that gradients cannot potentially leak local data. Although industry

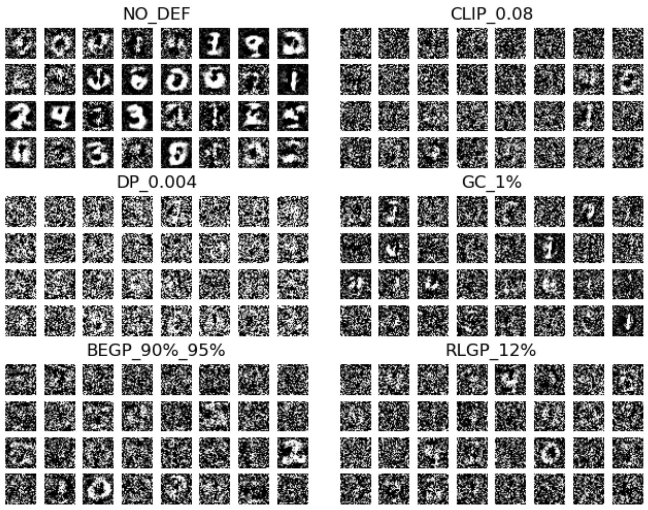


Fig. 11. Impact of DGL attack under several protection strategies on MNIST.

practices (e.g., homomorphic encryption, DP, and matrix transformation) are commonly used to address data leakage concerns, they often turn the aggregation process into a closed box operation, making it challenging to evaluate the contribution of each data provider. This is also why many FL incentive schemes do not take data leakage into account.

Fortunately, Zhu et al. [45] have proposed some straightforward and reliable defense schemes, including DP, gradient clipping (CLIP), and GC. Zhang et al. [46] identified the pivotal role of large gradients in the image reconstruction process. In addressing this concern, they introduced the relaxed large gradient pruning (RLGP). Integrating gradient pruning at both ends (BEGP) for comparative testing.

To evaluate the impact of gradient protection strategies on model convergence and collaborative evaluation, we utilized the IMDS scenario on MNIST. We fine-tuned the hyperparameters of these methods using the DGL attack to preserve crucial gradient information. Specifically, for CLIP, the ℓ_2-norm is set to 0.08. In the case of DP, it involves adding Gaussian noise with the ℓ_2-norm set at 0.004. GC is specified as 95%.

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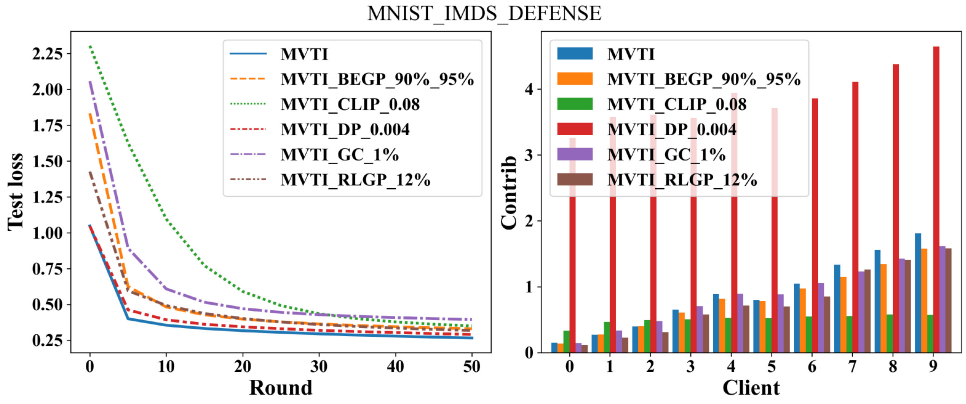


Fig. 12. Impact of above gradient protection strategies on model convergence and contribution evaluation of MVTI in the IMDS scenarios on MNIST.

For BEGP, weights are selected from the middle 90% to 95% of the absolute values of gradients. For RLGP, weights larger than 88% of the absolute values are pruned. The gradient is obtained from the optimized SGD with a batch size of 5 and a learning rate of 0.01, after 5 optimization rounds on 800 MNIST samples. Although not necessarily optimal, this approach allows for a realistic and practical comparison. The DGL attack defense effect is shown in Fig. 11.

In our experiments on gradient protection strategies, we assessed both the model convergence and the contribution evaluation of MVTI in the IMDS scenario on MNIST. As depicted in Fig. 12, DP has the least impact on model convergence, but poses challenges to the accuracy of incentive evaluation. In contrast, methods based on gradient pruning or compression exhibit relative superiority. The introduction of random noise by DP disrupts each update in model training, thereby slowing down the convergence speed. However, the stochastic nature of this noise results in unstable incentive evaluation, making it challenging to precisely measure each participant's contribution. On the other hand, gradient pruning or compression methods maintain the stability of model updates, enabling a more accurate assessment of incentive fairness and rationality. While pruning large gradients offers a significantly better defense effect than pruning small gradients, the latter excels in terms of model compression. Clipping small gradients helps reduce the model's parameters, thereby lowering computational and storage costs. In particular, in large-scale FL environments in AIoT, the advantages of model compression become more apparent, contributing to improved overall system efficiency and performance. Therefore, introducing BEGP as a strategy to balance communication efficiency, defense effectiveness, and incentive effectiveness proves to be a judicious choice.

VI. Conclusion and Future Work

In summary, this article introduces the MVTI, a pioneering mechanism for FL incentives in AIoT. MVTI enables active data requester participation in FL training, promoting personalized model construction for specific task requirements. Integrating a model bail and contribution assessment mechanism ensures fair redistribution of benefits, fostering collaboration among participants, even those with less quality data. To enhance security and reliability, this article employs SCs and IPFS, ensuring transparent and tamper-resistant execution. Extensive experiments comparing MVTI against state-of-the-art baselines underscore its effectiveness and superiority in real-world datasets. This article conducts gradient leakage attack experiments to assess compatibility with mainstream privacy protection strategies, revealing good alignment with GC. The SC-MVTI scheme emerges as a resilient solution that addresses challenges in FL incentive mechanisms, showcasing its potential to significantly improve the overall efficacy of FL systems within incentive frameworks. It is an interesting direction to improve the incentive effectiveness by rationalizing the model value assessment to rationalize the dynamically set model margins. Besides, we intend to design and analyze a more diverse and realistic bidding strategy of FL for AIoT.

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